

XU YUANJIAN(徐元健)

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EDUCATION

Hong Kong University of Science and Technology, Ph.D. in Fintech (Candidate) 2023.09–Present
Peking University, Master in Computer Science 2020.09–2023.06
Nankai University, Bachelor in Computer Science 2014.09–2018.06

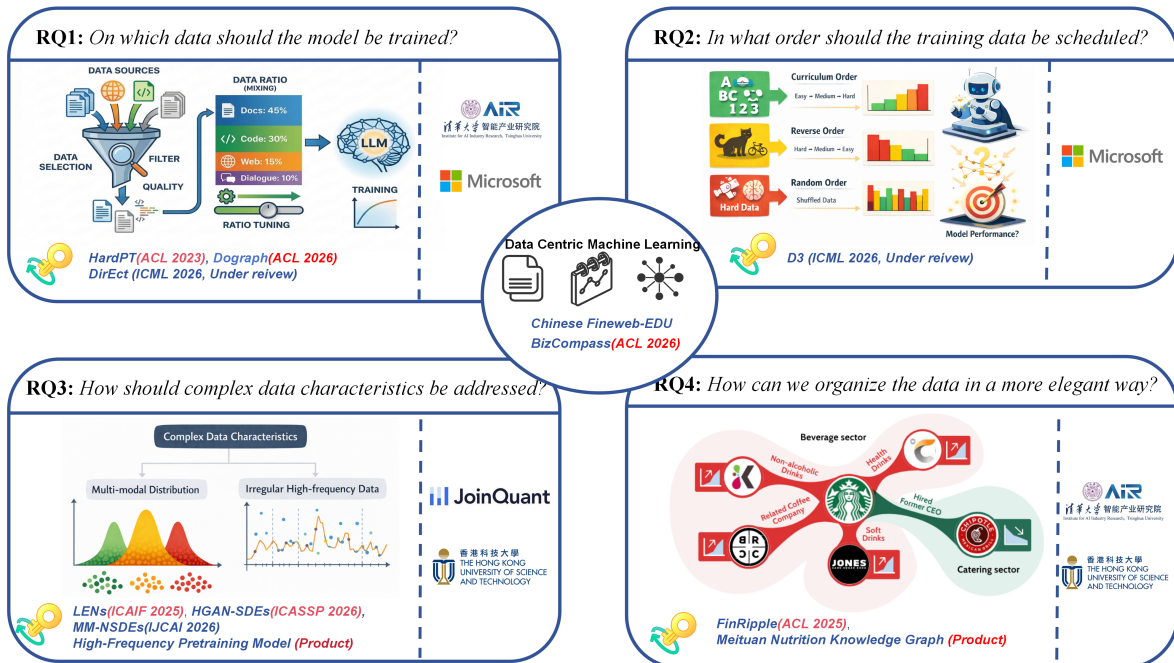
A BRIEF INTRODUCTION

I am a Ph.D. candidate in Computer Science at the Hong Kong University of Science and Technology, Guangzhou campus. I am supervised by Prof. Guang Zhang from HKUST-GZ.

My research interest revolves around **Data-centric Machine Learning**, with a primary focus on **LLMs**. Specifically, my work has systematically investigated the following four dimensions:

- **RQ1 (Data Selection)**: Identifying optimal data selection strategies for LLM pre-training.
- **RQ2 (Data Curriculum)**: Designing effective data scheduling to maximize model performance.
- **RQ3 (Data Representation)**: How to model *complex data characteristics* (e.g. multi-modal data)?
- **RQ4 (Data Orchestration)**: How to elegantly organize *heterogeneous data*?

To address these questions, I have published several papers at leading venues including **ACL 2023**, **ACL 2025**, **ACL 2026**, **ACM ICAIF 2025**, and **ICASSP 2026** (oral). Additionally, I have two manuscripts under review at **ICML 2026** (OA > 4) and one at **IJCAI 2026** (passed first round). I also serve as a reviewer for leading conferences such as **NeurIPS 2024/2025/2026**, **ICLR 2024/2025**, and **ICML 2026**.



SELECTED PUBLICATIONS (FIRST AUTHOR)

- [**ICML 2026**] **Yuanjian Xu**, et al. "D³: Dynamic Directional Graph-Constrained Data Scheduling for LLM Training." [CCF A] [CORE A*] (Incoming; All reviews are positive).
- [**ICML 2026**] **Yuanjian Xu**, et al. "Towards Efficient LLMs Annealing with Principled Sample Selection." [CCF A] [CORE A*] (Incoming; All reviews are positive).
- [**ACL 2026**] **Yuanjian Xu**, et al. "DoGraph: Towards Domain-Aware LLM Training with Graph-Guided Weighting." [CCF A] [CORE A*] (Incoming; All reviews are positive).

- [**ACL 2026**] **Yuanjian Xu**, et al. “BizCompass: Benchmarking the Reasoning Capabilities of LLMs in Business Knowledge and Applications.” [CCF A] [CORE A*] (Incoming; All reviews are positive).
- [**ACL 2025**] **Yuanjian Xu**, et al. “FinRipple: Aligning Large Language Models with Financial Market for Event Ripple Effect Awareness.” [CCF A] [CORE A*].
- [**ACL 2023**] **Yuanjian Xu**, et al. “Hard Sample Aware Prompt-Tuning.” [CCF A] [CORE A*].
- [**ICASSP 2026**] **Yuanjian Xu**, et al. “Hermite Discriminator for Neural SDEs.” [CCF B] [CORE A] (Oral).
- [**ICAIF 2025**] **Yuanjian Xu**, et al. “LENS: Large Pre-trained Transformer for Exploring Financial Time Series Regularities.” (Leading conference for AI in finance).

UNDER REVIEW & ONGOING WORKS (FIRST AUTHOR)

- [**IJCAI 2026**] **Yuanjian Xu**, et al. “State-Aware Neural Stochastic Differential Equations for Multi-Modal Dynamics.” [CCF A] [CORE A*] (Under review; passed first round).
- [**NeurIPS 2026**] **Yuanjian Xu**, et al. “FineWeb-Edu-Ultra: A Large-Scale Educational Dataset for Full-Cycle Foundation Model Training.” [CCF A] (Ongoing).
- [**NeurIPS 2026**] **Yuanjian Xu**, et al. “Data Correction via Logical Dependency Graph Reconstruction.” [CCF A] (Ongoing).
- [**NeurIPS 2026**] **Yuanjian Xu**, et al. “FinCorpus: A Large-Scale Corpus for Financial Foundation Models.” [CCF A] (Ongoing).

RESEARCH & TEACHING EXPERIENCE

Hong Kong University of Science and Technology, Teaching Assistant, Advanced Statistics

INTERNSHIP EXPERIENCE

AIR, Tsinghua University	Research Intern 2022.03–2023.03
Worked under the supervision of Prof. Zaiqing Nie. Contributed to the <i>Meituan Nutrition Knowledge Graph</i> construction. Investigated <i>hard sample problems</i> in NLP and proposed <i>HardPT</i> , published at <i>ACL 2023</i> .	
Microsoft Research Asia (MSRA)	Research Intern 2024.10–2025.02
Worked under the supervision of Dr. Zhong Li. Focused on <i>data selection</i> and <i>training order optimization</i> for large language models. Proposed the <i>D³ method</i> and an <i>annealing training framework</i> , both currently under submission.	
Joinquant (Billion-scale quantitative fund)	Research Intern 2025.02–2025.05
Worked under the supervision of PM Ruixiao. Developed <i>tick-level generative and representation models</i> for high-frequency trading. Addressed key challenges including <i>non-equally spaced data</i> and market randomness.	

SKILLS

Machine Learning: I am well-versed in modern generative models, LLMs, and machine learning.
Mathematical Foundation: I have a relative solid mathematical foundation. During my Ph.D. studies, I passed the qualifying exam in *optimization*, *advanced probability theory*, *advanced statistics*, and *stochastic analysis*.
Financial Theory: I also have exposure to financial theory, having passed *FRM Level 1* and studied *advanced microeconomics* and *advanced asset pricing*.

HONORS & AWARDS

• Award for Excellent Academic Excellence, Peking University	2021 Certificate No.: H2021000170320
• Air Star Plan, Tsinghua University, Institute for AI Industry Research (AIR)	2021
• Full Ph.D. Scholarship, Hong Kong University of Science and Technology	2023–Present
• Tomorrow Star Plan, Microsoft Research Intern	2025